

Properties of Limits

Thm. The limit of a function as $x \rightarrow a$, either DNE or it is unique.

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$$\begin{aligned} \text{If } \lim_{x \rightarrow a} f(x) &= l \\ \lim_{x \rightarrow a} f(x) &= m \\ \Rightarrow l &= m. \end{aligned}$$

Pf. First assume $\lim_{x \rightarrow a} f(x)$ exists

Suppose for a contradiction that $l \neq m$,

$$\lim_{x \rightarrow a} f(x) = l$$

$$\lim_{x \rightarrow a} f(x) = m$$

Let's write down the definition of limits in these cases:

$\forall \varepsilon > 0$, then $\exists \delta_1, \delta_2 > 0$, s.t.

$$\begin{aligned} \bullet \forall x \text{ with } 0 < |x - a| < \delta_1 \\ \Rightarrow |f(x) - l| < \varepsilon. \quad (1) \end{aligned}$$

$$\begin{aligned} \bullet \forall x \text{ w / } 0 < |x - a| < \delta_2 \\ \Rightarrow |f(x) - m| < \varepsilon \quad (2) \end{aligned}$$

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then for this δ , if $0 < |x - a| < \delta$

$\Rightarrow$ (1) (2) both true.

So for $0 < |x - a| < \delta$,

$$\begin{aligned} |m - l| &= |m - f(x) + f(x) - l| \\ &\leq |m - f(x)| + |f(x) - l| \\ &= |f(x) - m| + |f(x) - l| < \varepsilon + \varepsilon = 2\varepsilon \end{aligned}$$

If we take $\varepsilon < \frac{|l-m|}{2}$,

We have $|m-l| < 2\varepsilon < |l-m|$. Contradiction.



Thm. $\lim_{x \rightarrow a} f(x) = l \quad \lim_{x \rightarrow a} g(x) = m.$

then $\lim_{x \rightarrow a} (f+g)(x) = l+m.$

Roughwork Given $\varepsilon > 0$

want $|(f+g)(x) - (l+m)| < \varepsilon$ for x close to a . find δ .

$$|(f(x)-l) + (g(x)-m)| < |f(x)-l| + |g(x)-m|.$$

P.F. let $\varepsilon > 0$.

Then $\exists \delta_1, \delta_2 > 0$ s.t.

$$0 < |x-a| < \delta_1 \Rightarrow |f(x)-l| < \frac{\varepsilon}{2}$$

$$0 < |x-a| < \delta_2 \Rightarrow |g(x)-m| < \frac{\varepsilon}{2}$$

$$\text{let } \delta = \min\{\delta_1, \delta_2\}.$$

Then ① ② both hold at the same time

So now look at $\forall x$ s.t.

$$0 < |x-a| < \delta$$

$$\text{Then } |(f+g)(x) - (l+m)|$$

$$= |f(x) + g(x) - l - m|$$

$$= |(f(x)-l) + (g(x)-m)|$$

$$\leq |f(x)-l| + |g(x)-m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$



Thm. If $\lim_{x \rightarrow a} f(x) = l$, $\lim_{x \rightarrow a} g(x) = m$
then $\lim_{x \rightarrow a} (fg)(x) = lm$

Roughwork For $|x-a|$

Want $|(fg)(x) - lm| < \varepsilon$.

$$= |f(x)g(x) - lm|$$

$$= |f(x)g(x) - f(x)m + f(x)m - lm|$$

$$< |f(x)||g(x) - m| + m|f(x) - l|.$$

pf. Let $\varepsilon > 0$

Then $\exists \delta_1, \delta_2$ s.t.

- If $0 < |x-a| < \delta_1$, then $|f(x) - l| < \frac{\varepsilon}{2m}$.
 - If $0 < |x-a| < \delta_2$, then $|g(x) - m| < \varepsilon_0$.
- } (X)

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then $|(fg)(x) - lm|$

$$= |f(x)g(x) - lm|$$

$$= |f(x)g(x) - f(x)m + f(x)m - lm|$$

$$\leq |f(x)(g(x) - m)| + |(f(x) - l)m|$$

$$= |f(x)||g(x) - m| + |f(x) - l||m|.$$

$$< |f(x)|\varepsilon_0 + \frac{\varepsilon}{2|m|}|m|.$$

Declare that $\varepsilon < 1$.

Then from the 1st ineq of (X)

$$|f(x) - l| < \frac{1}{2|m|}$$

Hence, by triangle inequality,

$$\begin{aligned}|f(x)| &= |f(x) - l + l| \\ &\leq |f(x) - l| + |l| \\ &\leq \frac{1}{2|m|} + |l|.\end{aligned}$$

$$\text{Define } \varepsilon_0 = \frac{\varepsilon}{2\left(\frac{1}{2|m|} + |l|\right)}$$

$$\begin{aligned}\text{Then, } |(fg)(x) - lm| &\leq |f(x)| |g(x) - m| + |f(x) - l| |m| \\ &< \left(\frac{1}{2|m|} + |l|\right) \varepsilon_0 + \frac{\varepsilon}{2|m|} |m| \\ &= \left(\frac{1}{2|m|} + |l|\right) \frac{\varepsilon}{2\left(\frac{1}{2|m|} + |l|\right)} + \frac{\varepsilon}{2} \\ &= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon\end{aligned}$$

